

Computational Modelling of DWR Re-Proofer Drying on Woven Textiles

Colloidal Stratification, Crown Starvation, and
Process Window Prediction

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Portfolio project — qualitative computational model; no experimental DWR measurements were performed.

Abstract. A physics-based computational model is presented for the drying of a durable water repellent (DWR) wash-in re-treatment on a woven textile. The model couples a 1D binary-colloid evaporation solver — derived from the Routh/Russel film-Péclet framework — to an idealised plain-weave unit cell, and outputs a predicted water-beading performance index. The central finding is that capillary pooling draws the dispersion into the inter-yarn interstices, producing locally thick residual films with a high film Péclet number; the functional polymer is therefore swept toward the drying surface and buried rather than forming a usable repellent film. The exposed yarn crowns — the surfaces a raindrop actually rests on — are consequently starved of polymer. A 2D process-window map identifies the combination of polymer loading and drying rate that mitigates this crown-starvation mechanism.

1 Introduction and Motivation

Consumer DWR wash-in re-treatments are aqueous colloidal dispersions: a functional water-repellent polymer (large particles, radius $r_L \sim 150$ nm) co-suspended with a smaller carrier/surfactant phase ($r_S \sim 40$ nm). After a garment is washed in the treatment and removed, the water evaporates. As the air-liquid interface recedes it advects particles toward the drying substrate; diffusion opposes this concentration. The competition is governed by the film Péclet number,

$$\text{Pe} = \frac{v_{\text{evap}} H_0}{D}, \quad (1)$$

where v_{evap} is the interface recession velocity, H_0 the initial film thickness, and D the Stokes-Einstein diffusivity $k_B T / (6\pi\mu r)$. Because $D \propto 1/r$, the large functional polymer has a smaller diffusivity and a correspondingly higher Pe, making it more susceptible to burial.

In a woven textile this drying physics operates on a non-uniform film: capillary pressure preferentially draws the dispersion into the high-curvature inter-yarn interstices, leaving a thicker residual film there than on the exposed yarn crowns. The locally thicker film means a locally higher Pe — exactly where you *least* want the polymer to be buried.

This project transplants the 1D colloidal-film stratification physics from the author’s MEng dissertation (flat binary-colloid films) onto a woven textile, coupling it to a spatial deposition model and a predicted water-beading performance index.

2 Model

2.1 1D binary-colloid drying solver

The DWR film on the textile surface is modelled as a locally 1D evaporating binary film. The governing equation uses the conservative areal-density variable $q = h\phi$ on a mapped coordinate $x = z/h(t)$ that transforms the shrinking physical domain $[0, h(t)]$ onto a fixed grid $[0, 1]$:

$$\frac{\partial q}{\partial t} = \frac{\partial}{\partial x} \left[\frac{D}{h^2} \frac{\partial q}{\partial x} + \frac{\dot{h}}{h} x q \right]. \quad (2)$$

The first term is Fickian diffusion with a mapped coefficient D/h^2 (which diverges as $h \rightarrow 0$); the second is the moving-frame advective concentration driving stratification. The transformation was derived and symbolically verified with `sympy`; an earlier formulation that dropped the advection term was caught during this check.

The numerical scheme uses:

- Conservative finite-volume discretisation on fixed cell centres — $\int q \, dx$ is conserved to $\lesssim 10^{-12}$ (verified over 12 Péclet–step combinations).
- Backward-Euler implicit time stepping — unconditionally stable, so the D/h^2 stiffness never limits the step size.
- Stokes–Einstein diffusivity for each species.

2.2 Plain-weave unit cell and performance model

An idealised plain-weave unit cell is constructed from two orthogonal families of sinusoidal yarns in anti-phase. Each surface point is assigned (a) a local height (crown vs. valley) and (b) a capillary residual-film thickness, using a smooth monotone map of $(1 - h)$ so that deep inter-yarn valleys pool the most liquid (Fig. 3).

The pipeline from geometry to predicted performance is:

1. **Local drying.** The 1D solver is evaluated at a small set of representative film thicknesses and interpolated. Each location’s local Pe is set by its local film thickness.
2. **Effective coverage.** The usable-surface fraction (fraction of functional polymer ending up at the *air* side of the dried deposit rather than buried) is combined with the deposited mass per unit area to give an effective DWR areal coverage in $[0, 1]$.
3. **Contact angle.** A Cassie–Baxter mixing rule between the fully-coated ($\theta_{\text{DWR}} = 105^\circ$) and bare-yarn ($\theta_{\text{bare}} = 65^\circ$) states yields the local effective contact angle.
4. **Beading index.** A crown-weighted spatial average of a sigmoid beading score (centred at 90° , the wetting–beading transition) produces a single index in $[0, 1]$. This is a *relative* metric for ranking processing conditions, not an absolute contact-angle prediction.

3 Results

3.1 Drying profiles and stratification

Fig. 1 shows the functional-polymer distribution at low and high Pe. At low Pe (slow drying or thin film) the deposit is near-uniform; at high Pe the large species is swept toward the film surface during drying and then buried against the substrate as the film collapses — exactly the condition that degrades beading. Fig. 2 confirms that the stratification index rises monotonically with Pe, consistent with the Fortini et al. binary-stratification prediction.

Drying-induced redistribution of functional polymer through the residual film

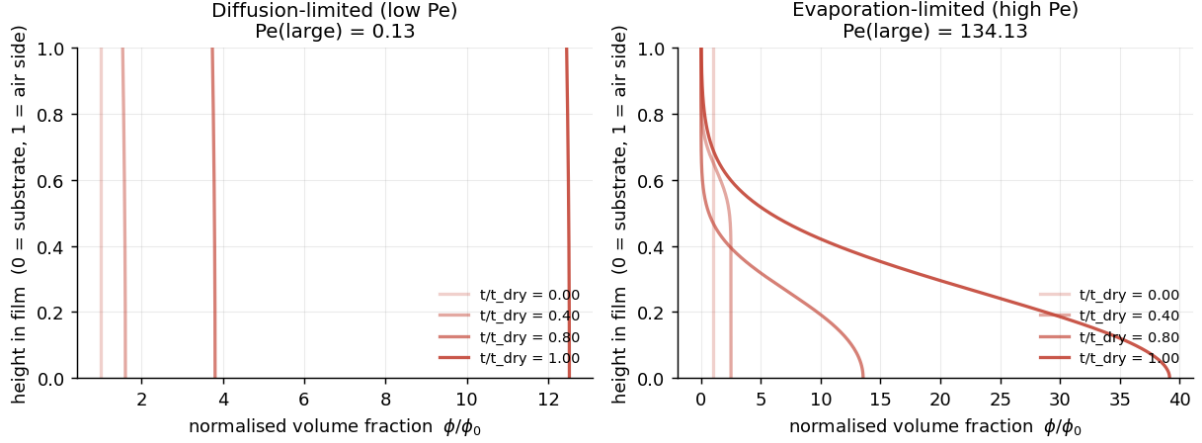


Figure 1. Volume-fraction profiles through the film depth at end of drying for low Pe (left) and high Pe (right). At low Pe diffusion maintains a near-uniform deposit; at high Pe the large species (functional polymer) is strongly concentrated at the top of the film and then buried against the substrate as the film collapses.

3.2 Weave geometry and the pooling map

The plain-weave unit cell (Fig. 3) shows that capillary pooling produces a film-thickness ratio of $\approx 4.5\times$ between interstices and crowns. Because $Pe \propto H_0$, interstices have correspondingly higher local Pe, making them the strongest burial zones.

3.3 Crown starvation

Fig. 4 is the central result. At moderate polymer loading the contact angle is *higher in the valleys* than on the crowns ($\approx 94^\circ$ vs. $\approx 86^\circ$). The thick-film interstices accumulate total polymer but bury it; the thinner-film crowns dry more uniformly but receive less total material. The net outcome is crown starvation — a known failure mode of consumer re-treatment that the model captures and quantifies.

3.4 Process window

Fig. 5 maps the beading index across combinations of polymer loading and drying rate. Three regimes are evident: (i) too little polymer — insufficient deposit regardless of drying rate; (ii) fast/hot drying — high Pe everywhere drives burial, suppressing usable coverage even at high loading; (iii) the process sweet spot — moderate-to-high loading ($\phi_0 \gtrsim 0.07$) combined with slow, cool drying (low evaporation rate), giving beading index > 0.75 .

4 Implications for Materials Development and QA

Formulation. Polymer loading is the most robust lever: increasing ϕ_0 from 0.045 to 0.09 raises the beading index from ≈ 0.35 to ≈ 0.75 at slow drying rates. However, loading alone cannot compensate for very fast drying — the fundamental burial mechanism sets a ceiling.

Process control. The model identifies drying rate as the second critical parameter. A re-proofer dried in a hot tumble dryer (high v_{evap} , high Pe) consistently underperforms the same formulation dried at ambient temperature, even at the same polymer loading. Consumer instructions to “dry naturally or on a low-heat setting” are physically well-motivated by this mechanism.

Weave structure. Fabrics with deeper inter-yarn channelling (higher crimp, tighter sett) pool more

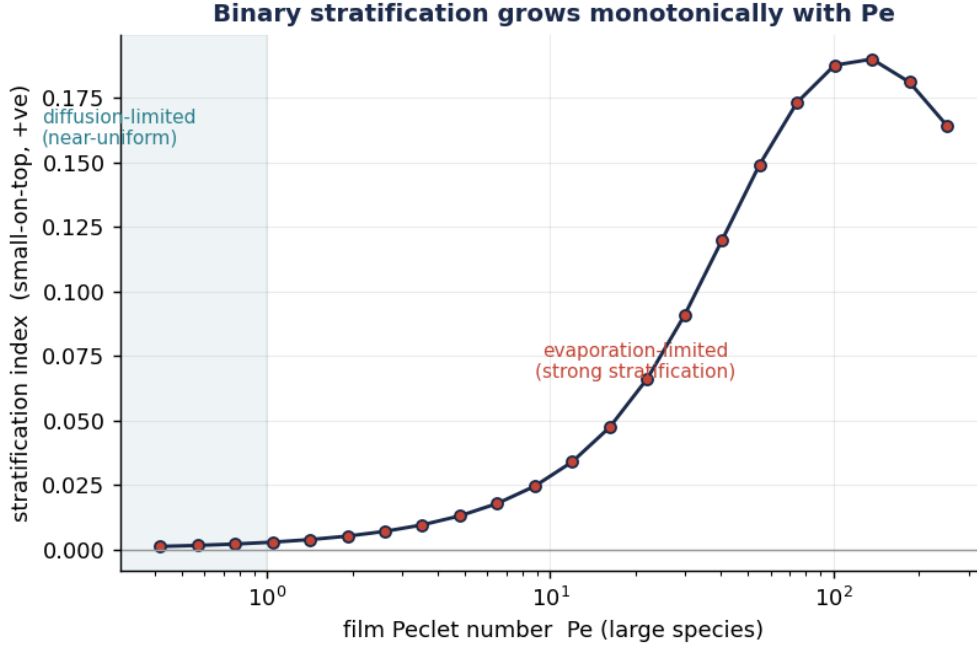


Figure 2. Stratification index (difference in centre-of-mass height between small and large species, normalised by final film height) as a function of Pe . The index rises monotonically from ≈ 0 at low Pe to > 0.18 at high Pe , reproducing the qualitative behaviour of the Routh/Russel and Fortini binary-stratification frameworks.

dispersion in the interstices, increasing the crown-starvation penalty. The model’s crimp parameter directly quantifies this trade-off and could guide fabric specification decisions for re-treatability.

QA indicator. The predicted crown-to-valley contact-angle ratio is a directly measurable quantity (e.g. by confocal microscopy on cross-sections, or by goniometry at defined fabric orientations) that would provide experimental grounding for the relative beading index.

5 Limitations

- **1D drying physics.** Lateral capillary flow during drying (which drives the pooling) is parameterised rather than solved; a coupled thin-film lubrication model would make the film-thickness map predictive rather than assumed.
- **Idealised geometry.** The sinusoidal plain-weave unit cell captures topology but not fibre-scale roughness, twist, or crimp geometry of real yarns. A fibre-resolved micro-CT model is the natural extension.
- **Constant evaporative flux.** The falling-rate-free period only; real garment drying involves a surface-dry transition and through-thickness moisture gradients.
- **No film-formation kinetics.** The model predicts polymer deposition location, not whether it forms a continuous, defect-free DWR film (which requires exceeding the minimum film-forming temperature).
- **Relative beading index.** The $[0, 1]$ index correctly *ranks* processing conditions but is not calibrated to absolute contact-angle data. One spray/bead test at two drying temperatures would anchor it.

Idealised woven unit cell: geometry sets where the re-proofer pools

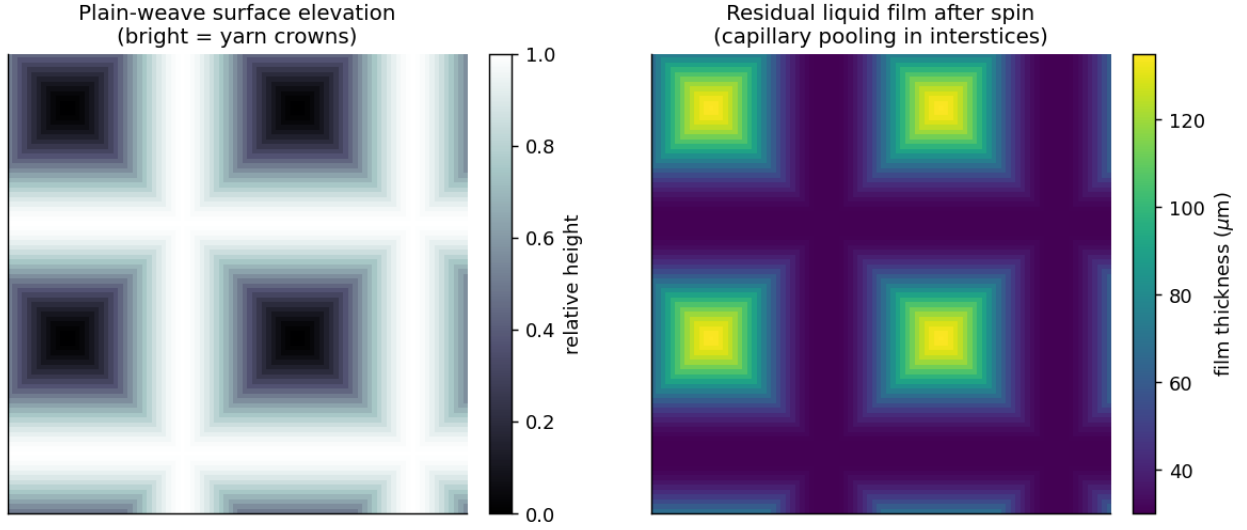


Figure 3. Plain-weave unit-cell geometry. *Left:* surface elevation (crowns bright, valleys dark). *Centre:* capillary residual-film thickness map — film depth increases from $\approx 30\ \mu\text{m}$ on crowns to $\approx 135\ \mu\text{m}$ in interstices. *Right:* local film Péclet number; interstices are the highest-Pe zones.

6 Conclusions

A physics-based model has been built and verified against the known theoretical behaviour of colloidal-film stratification. The central finding — that capillary pooling in the inter-yarn interstices drives a high-Péclet burial of functional polymer, starving the exposed yarn crowns — is mechanistically clear, quantified, and reproducible in a locked computational test. The process-window output provides an actionable, two-parameter design guide: adequate polymer loading ($\phi_0 \gtrsim 0.07$) combined with slow, cool drying is both necessary and sufficient for good predicted beading performance. The immediate next step to ground this model is a two-temperature spray/bead test, which would anchor the relative beading index to absolute contact-angle measurements and close the loop between computation and experiment. The model is openly available with full source code, 12 passing verification tests, and an interactive dashboard for real-time parameter exploration.

Software. Python 3 with NumPy, SciPy, Matplotlib, and SymPy. All results are predictions of a computational model, verified against the known theoretical behaviour of the colloidal-film stratification literature. No experimental DWR measurements were performed.

Scientific basis. Routh & Russel (evaporation-driven colloidal-film drying, film Péclet framework); Fortini et al. (binary stratification, small-on-top mechanism); Cassie & Baxter (wetting on heterogeneous surfaces); Stokes & Einstein (particle diffusivity scaling).

Capillary pooling starves the yarn crowns -- the surfaces a droplet actually rests on

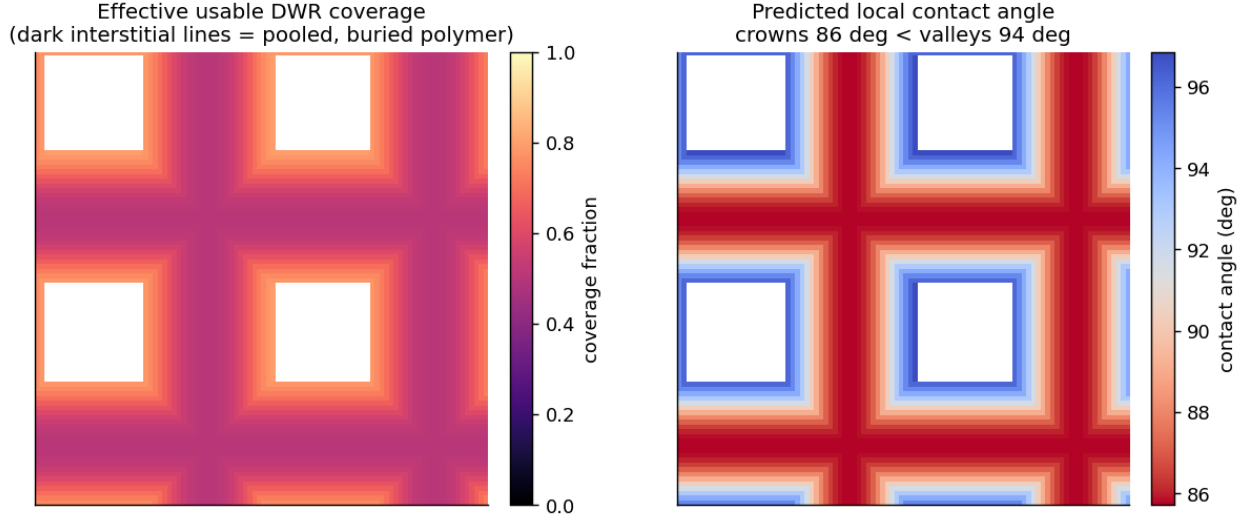


Figure 4. Spatial deposition results at moderate polymer loading ($\phi_0 = 0.045$). *Left:* effective DWR coverage. *Right:* predicted contact-angle map (Cassie-Baxter). Valley contact angles ($\approx 94^\circ$) exceed crown contact angles ($\approx 86^\circ$) — the crowns are starved of functional polymer despite carrying the full load of raindrop contact.

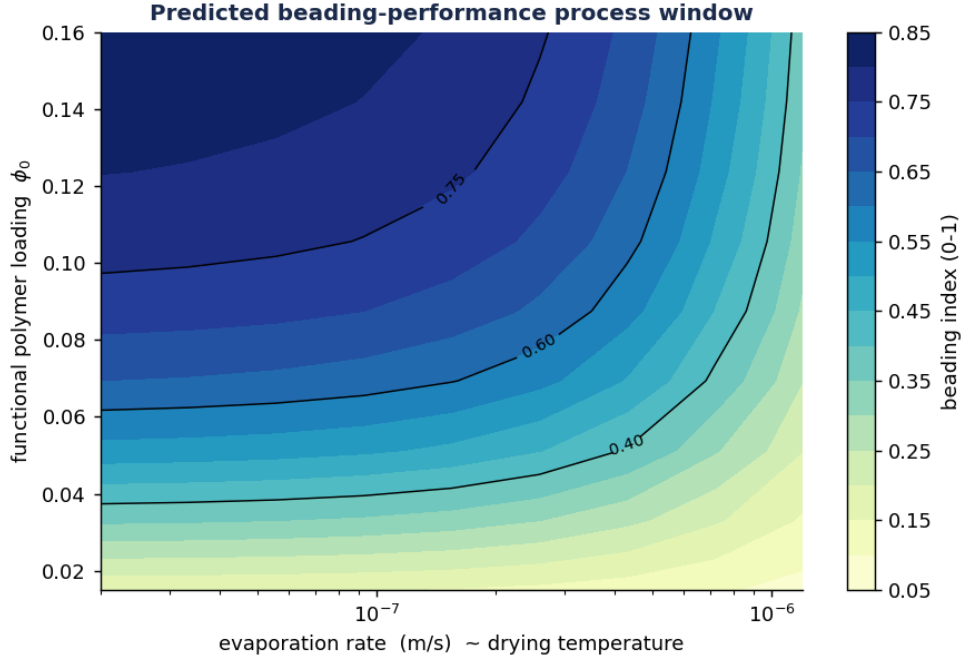


Figure 5. Beading index over the 2D process space of polymer loading ϕ_0 and evaporation rate. High index (bright) requires adequate loading *and* slow drying. Dashed contour marks the 0.75 threshold.